

Problem 3

Problem 3

Problem 1 Suppose that $f(5) = 7$, $f'(5) = 8$, $g(5) = 3$, and $g'(5) = -4$. Find:

(a) $(fg)'(5)$.

Explanation.

$$\begin{aligned}(fg)'(5) &= (f'(5) \cdot g(5)) + (f(5) \cdot g'(5)) \\ &= (8)(3) + (7)(-4) \\ &= 24 - 28 = -4.\end{aligned}$$

(b) $\left[\frac{d}{dx} \left(\frac{f}{g} \right) \right]_{x=5}$

Explanation.

$$\begin{aligned}\left[\frac{d}{dx} \left(\frac{f}{g} \right) \right] (5) &= \frac{(g(5) \cdot \left[\frac{d}{dx} f \right] (5)) - (f(5) \cdot \left[\frac{d}{dx} g \right] (5))}{(g(5))^2} \\ &= \frac{(3)(8) - (7)(-4)}{3^2} \\ &= \frac{24 + 28}{9} = \frac{52}{9}.\end{aligned}$$

(c) $\left(\frac{g}{f} \right)' (5)$

Explanation.

$$\begin{aligned}\left(\frac{g}{f} \right)' (5) &= \frac{(f(5) \cdot g'(5)) - (g(5) \cdot f'(5))}{(f(5))^2} \\ &= \frac{(7)(-4) - (3)(8)}{7^2} \\ &= \frac{-28 - 24}{49} = -\frac{52}{49}.\end{aligned}$$

(d) $\left[\frac{d}{dx} \left(\frac{g(x)}{x+2} \right) \right]_{x=5}$

Explanation.

$$\begin{aligned} \left[\frac{d}{dx} \left(\frac{g(x)}{x+2} \right) \right]_{x=5} &= \left[\frac{(x+2) \cdot g'(x) - (g(x) \cdot (x+2)')}{(x+2)^2} \right]_{x=5} \\ &= \frac{(5+2)(-4) - (3)(1)}{(5+2)^2} \\ &= \frac{(7)(-4) - (3)(1)}{7^2} \\ &= \frac{-28 - 3}{49} = -\frac{31}{49} \end{aligned}$$

(e) $\left[\frac{d}{dx} \left(\frac{xf(x)}{g(x)} \right) \right]_{x=5}$

Explanation.

$$\begin{aligned} \left[\frac{d}{dx} \left(\frac{xf(x)}{g(x)} \right) \right]_{x=5} &= \left[\frac{(((xf(x))' \cdot g(x)) - (g'(x) \cdot (xf(x))))}{(g(x))^2} \right]_{x=5} \\ &= \left[\frac{((x)' \cdot f(x) + x \cdot f'(x)) \cdot g(x) - (g'(x) \cdot (xf(x)))}{(g(x))^2} \right]_{x=5} \\ &= \frac{(1 \cdot 7 + 5 \cdot 8)(3) - (-4)(5)(7)}{(3)^2} \\ &= \frac{141 + 140}{9} \\ &= \frac{281}{9} \end{aligned}$$

(f) $\lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5}$. **EXPLAIN.**

Explanation. We first note that this limit has form $\frac{0}{0}$, but with no formula for $g(x)$ there is no simplification we can perform. Notice that if we set $w(x) = x^2 g(x)$, then $w(5) = (5)^2 g(5) = 75$. This limit is the definition of the derivative, $w'(5)$.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5} &= w'(5) \\ &= \left[\frac{d}{dx} (x^2 g(x)) \right]_{x=5} \\ &= \left[2xg(x) + x^2 g'(x) \right]_{x=5} \\ &= 2(5)g(5) + (5)^2 g'(5) \\ &= -70 \end{aligned}$$

$\lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5} = -70$ by the product rule since this is the definition of the derivative of $x^2 g(x)$ at $x = 5$.

(g) $\lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5}$. **EXPLAIN.**

Explanation. We first note that this limit has form $\frac{0}{0}$. Notice that if we set $k(x) = \frac{x+2}{f(x)}$, then $k(5) = \frac{5+2}{f(5)} = 1$. This limit is the definition of the derivative, $k'(5)$.

$$\begin{aligned} \lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5} &= k'(5) \\ &= \left[\frac{d}{dx} \left(\frac{x+2}{f(x)} \right) \right]_{x=5} \\ &= \left[\frac{f(x) - (x+2)f'(x)}{(f(x))^2} \right]_{x=5} \\ &= \frac{f(5) - (5+2)f'(5)}{(f(5))^2} \\ &= \frac{7 - 7(8)}{(7)^2} \\ &= -1 \end{aligned}$$

$\lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5} = -1$ by the quotient rule since this is the definition of the derivative of $\frac{x+2}{f(x)}$ at $x = 5$.